

12.1 12.2

12.1

We have an exchange operator, and can't allow symmetry under exchange (and have to have a phase) for some arbitrary reason, somehow spin comes in and we can allow bosons in the same state but not fermions but with no justification and somehow this is a deep truth of the universe.

I hate this textbook. Nothing is tied to a physical operation, and its super hand wavy while also claiming immense power to lord over the world. I understand how quantum is useful and have seen its effects, but gosh dang this is so unmotivated and feels so arbitrary.

12.2

Somehow we asserted that the total spin state for fermions has to be zero.

integrals to find the ground state energy

higher energy things

no clue whats happening

I hate this and feel no motivation to read the textbook, its just random factoids pulled from nowhere no motivation or easy to follow structure, and I would not have been able to derive anything on my own, nor use it at all.